ANALYTICAL CUMULATIVE EXAM

December 5, 1992

ANSWER ANY THREE OF THE FOLLOWING:

1. Given the following information

$$Cr_2O_7^{2-} + 14\ H^+ + 6\ e^- \rightarrow 2Cr^{+3} + 7H_2O \qquad E° = 1.33 \text{ volts}$$

$$Sn^{4+} + 2\ e^- \rightarrow Sn^{2+} \qquad E° = 0.15 \text{ volts}$$

$$2.303\ \frac{RT}{F} = 0.0591$$

Consider the cell

$Pt|Cr_2O_7^{2-}$ (0.05M), Cr^{+3} (0.01M), H^+ (pH = 2.00)$||Sn^{4+}$ (0.10M), Sn^{2+} (0.02M)$|Pt$

a. Write a balanced equation for the overall cell reaction.
b. Calculate E° for the cell.
c. Calculate E for the cell.

2. A 6.044 g sample containing KI, KBr and inert substances was dissolved in water and diluted to 500.0 mL in a volumetric flask. Treatment of a 50.00 mL aliquot with Br_2 resulted in conversion of I^- to IO_3^-. The excess Br_2 was eliminated by boiling, following which the solution was acidified and treated with an excess of KI. The liberated I_2 was titrated with 41.7 mL of 0.0489 N $Na_2S_2O_3$.

The Br_2 and I_2 produced by oxidation of a 10.00 mL aliquot with strongly acidic $Cr_2O_7^{2-}$ were distilled into a flask containing a concentrated solution of KI. At the conclusion of the distillation the I_2 in the receiving vessel was titrated with 30.57 mL of the $Na_2S_2O_3$ solution.

Calculate the percentages of KI and KBr in the sample.

3. Under a particular set of conditions, exactly one half of the metal ion in an aqueous sample was extracted into an equal volume of a non-miscible organic solvent containing a chelating agent. Calculate the number of extractions, each with an equal volume of fresh organic solvent, needed to remove at least 90.0% of the metal ion from the water.

4. Consider the titration of 0.10 M phosphoric acid (pK_1 = 2.1, pK_2 = 7.2 and pK_3 = 12.3) with 0.10 M NaOH. Under the conditions that one-half of the H_3PO_4 has been converted to $H_2PO_4^-$, what is the concentration of PO_4^{3-}? At what pH will 90.0% of the acid be converted to PO_4^{3-} and what volume of the base will be required to reach this value?